Preserving PET Quantification in Longitudinal Whole-Body Registration

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Abstract. Longitudinal PSMA PET/CT registration has to align whole-body scans that differ in field of view, positioning and bowel and bladder filling, while preserving the PET-derived biomarkers that the scans are acquired to measure. Accuracy and preservation pull against each other: a deformation that maximises organ overlap is free to compress or expand a lesion, changing exactly the quantity a clinician reads. We adapt the diffeomorphic, coarse-to-fine LapIRN architecture to this task for the Learn2Reg 2026 PSMAReg challenge, and add two anatomically motivated constraints to its objective. The first is a set of biomarker terms that pin the metabolic tumour volume and total lesion glycolysis under the estimated transformation, including a Jacobian-based surrogate that supplies gradients throughout the lesion interior rather than only at its boundary. The second is a bone-rigidity penalty formulated as a per-structure rigid fit: classical derivative-based rigidity terms read a finite-difference stencil that, on ribs and cortical bone at whole-body resolution, is dominated by surrounding soft tissue, whereas a closed-form rigid fit per skeletal label reads nothing outside it. The same objective is descended per case at test time by instance optimisation. [TODO: results sentence: DSC, HD95, MTV/TLG bias and NDV against the baselines, plus the leaderboard placement]

Keywords: Deformable image registration · PSMA PET/CT · Longitudinal imaging · Biomarker preservation · Learn2Reg.

1 Introduction

Prostate-specific membrane antigen (PSMA) PET/CT has become a central imaging modality for staging, treatment planning, radiopharmaceutical therapy, and post-therapy response assessment in prostate cancer [6]. In these settings, patients often undergo multiple scans over time, comprising a baseline scan before therapy and one or more follow-up scans afterwards; accurate baseline-to-follow-up registration can support comparison of disease burden, localisation of uptake changes, assessment of lesion progression or response, dose accumulation, and quantitative longitudinal analysis [14]. The challenge organisers identify several factors that make establishing this correspondence by deformable image registration difficult in routine whole-body data [14]: baseline and follow-up scans differ substantially in field of view, patient positioning, respiratory state, bladder and bowel filling, and body weight; CT and PET encode complementary but appearance-dissimilar information; and the registration must preserve the quantitative PET signal used for response assessment while remaining physically plausible structures such as the skeleton. The Learn2Reg 2026 PSMAReg challenge [10, 9, 11] formalises this problem as the estimation of a dense displacement field that warps each follow-up (moving) scan to its baseline (fixed) scan, evaluated jointly on organ overlap, boundary distance, deformation regularity, and preservation of PET-derived tumour biomarkers.

Deformable registration has traditionally been posed as an iterative optimisation over a regularised similarity objective, with widely used instances including the diffeomorphic Demons [27], SyN [1], and B-spline [24] schemes. While accurate, these methods are slow at whole-body resolution. Learning-based methods instead amortise the optimisation over a training set into the network weights, replacing per-pair iterative optimisation with a single forward pass at inference. VoxelMorph established the unsupervised paradigm [3], and subsequent work introduced diffeomorphic parameterisations through stationary velocity fields [5], transformer backbones such as TransMorph [4], and multi-resolution architectures such as LapIRN [20]. Among the latter, LapIRN uses a Laplacian pyramid of coarse-to-fine sub-networks to recover large deformations while keeping the transformation smooth and invertible, and has been a strong performer across previous Learn2Reg editions [10, 9]. A recurring limitation of purely intensity-driven losses is that they permit non-physical folding. Anatomy-aware regularisation that penalises non-rigid deformation, e.g. of skeletal structures [16, 23, 25, 19], is therefore particularly relevant for whole-body PET/CT, where the skeleton is a frequent site of PSMA-avid disease.

In this work, we adapt LapIRN to longitudinal whole-body PSMA PET/CT for the PSMAReg task.

Our contributions are:

- an adaptation of the diffeomorphic, coarse-to-fine LapIRN architecture to longitudinal whole-body PSMA PET/CT, combining an affine pre-registration with a three-level pyramid and test-time instance optimisation;
- a set of biomarker-preserving losses that constrain the PET-derived MTV and TLG under the estimated deformation, including a Jacobian-based surrogate that supplies dense gradients throughout the lesion interior;
- a per-structure rigid-fit bone-rigidity penalty
- a score-driven checkpoint and submission selection procedure that mirrors the challenge ranking. [TODO: finalise once the model is fixed]

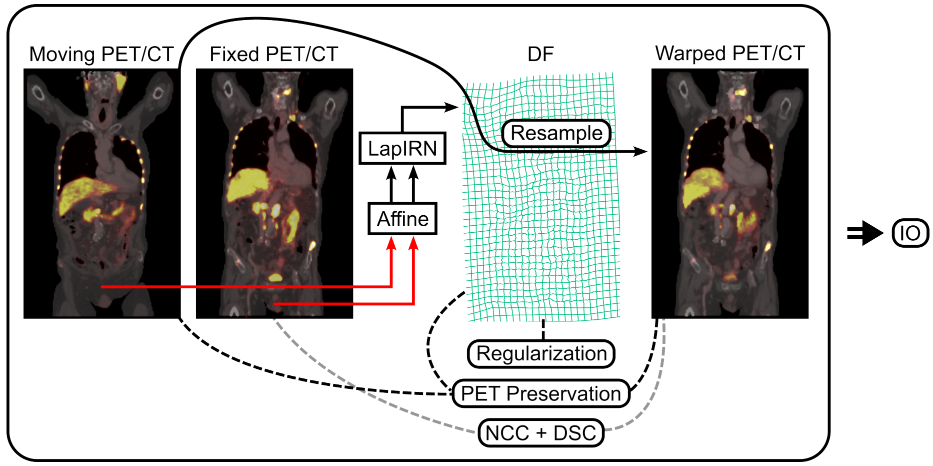

Fig. 1. Overview of the proposed registration pipeline. The moving (follow-up) and fixed (baseline) PET/CT are first coarsely aligned by an affine pre-registration (red), and the affinely aligned pair is passed to the coarse-to-fine LapIRN network, which predicts a dense displacement field (DF, green grid) used to resample the moving scan into the fixed space (warped PET/CT). Training is driven by three groups of objectives: deformation regularisation on the DF, comprising a global smoothness and folding penalty and a per-structure bone-rigidity term that constrains the DF inside skeletal labels and thereby also acts as an anatomical surrogate for biomarker preservation, since a large fraction of the lesion volume is intraosseous; PET-biomarker preservation terms (MTV, TLG and Jacobian inside the tumour mask) comparing the moving and warped lesions (black dashed); and image- and label-based similarity, NCC and DSC, between the fixed and warped scans (grey dashed). At test time, the same objectives are descended per pair by instance optimisation (IO).

The proposed registration pipeline with affine pre-registration, LapIRN-based registration, losses and instance optimization (IO) is shown in Fig. 1.

2 Methods

2.1 Data and Preprocessing

We used the PSMAReg dataset of the Learn2Reg 2026 challenge [14], a longitudinal whole-body PSMA PET/CT cohort adapted from the manually annotated autoPET data of [11]. Each paired subject provides a baseline scan and one or more follow-up scans, each with a CT and a PSMA PET channel; the follow-up is registered to the baseline, i.e. the baseline acts as the fixed and the follow-up as the moving image.

The challenge training split contains 146 paired and 212 unpaired subjects; the unpaired single-time-point cases were not used in this work. We divided the paired subjects patient-wise into 117 training and 29 internal validation subjects.

Registration pairs are formed within a subject and always point from the temporally later to the earlier scan. For training we used all such ordered time-point combinations, including follow-up-to-follow-up pairs, which yields 239 pairs from the 117 subjects and acts as a mild augmentation of the 146 baseline-anchored pairs. For our internal validation subjects we kept only baseline-anchored pairs, i.e. every follow-up registered to its baseline, matching the pairing used at evaluation time and giving 39 pairs from the 29 subjects. The challenge validation split of 20 paired subjects, which is the split scored on the public validation leaderboard, served as our test set; the organisers' own held-out test set ($\sim$200 pairs of private JHU data) is scored only by them.

For the training split we used the labels released by the organisers: CT organ labels generated automatically with TotalSegmentator [28] and PET lesion labels manually annotated by radiologists. Since no labels are released for the challenge validation split, we generated our own for our test set with TotalSegmentator for the CT organs and the winning autoPET III submission [22] for the PET lesions.

All images are provided already resampled to an isotropic in-plane spacing of $2.7344 \times 2.7344 \times 3.27$ mm and cropped to a fixed size of $192 \times 192 \times 288$ voxels, so no further geometric preprocessing was applied. We normalised CT by clipping to a fixed window of $[-1000, 1500]$ HU and PET by clipping the SUV to $[0, 20]$, each rescaled to $[0, 1]$.

2.2 Registration Pipeline

The pipeline consists of an affine pre-registration followed by a deformable LapIRN stage and, at test time, instance optimisation (Fig. 1). Throughout, I denotes the moving PET image, M its lesion mask, φ_{aff} the affine pre-registration, φ_θ the displacement field predicted by the network, and

$$\varphi = \varphi_{\text{aff}} \circ \varphi_\theta \tag{1}$$

the total transformation mapping the moving into the fixed image; W denotes the lesion region in the fixed frame, i.e. the image of M under φ.

Affine pre-registration. Baseline and follow-up scans of the same subject differ in field of view, table position and patient posture by far more than a locally regularised network can absorb, so we first remove the global component with a separate affine stage. For each pair we estimate an affine transform with ANTs [2] on the CT channel alone, at half resolution and with a Mattes mutual-information metric, after masking both volumes to the body and windowing them to $[-1000, 1000]$ HU. The result is converted into a dense voxel displacement field, cached per pair, and applied to the moving scan before the network sees it; augmentations are applied consistently to the cached field. Note that an affine transform is in general not volume-preserving, which is why all biomarker terms of Sec. 2.3 are evaluated on the composed transform (1) rather than on φ_θ alone.

Deformable registration. For the deformable stage we use LapIRN [20], a Laplacian-pyramid network whose three sub-networks operate at a quarter, a half and the full resolution of the $192 \times 192 \times 288$ volumes. Each level predicts a stationary velocity field that is added to the upsampled velocity of the coarser level; the sum is exponentiated by scaling and squaring to give a diffeomorphic φ_θ, so large deformations are recovered coarse-to-fine while the transformation stays smooth and invertible. We adapted the architecture to the multi-modal, whole-body setting: each sub-network takes four input channels, the CT and PSMA PET of the affinely aligned moving scan and of the fixed scan, and uses 24 initial feature channels with five residual blocks per level. Image similarity is computed on CT only, since PET uptake is not conserved between time points and would otherwise drive the deformation towards matching disease rather than anatomy; the PET channel enters the network as input and the biomarker terms of Sec. 2.3 instead. The levels are trained in sequence, each initialised from the previous one, which is kept frozen for the first ten epochs and fine-tuned jointly afterwards.

2.3 Loss Functions

The full objective combines image similarity, weak label supervision, the PET-biomarker terms, the bone-rigidity penalty and two deformation regularisers:

$$\begin{aligned}\mathcal{L} = {} & \lambda_{\mathrm{NCC}}\,\mathcal{L}_{\mathrm{NCC}} + \lambda_{\mathrm{DSC}}\,\mathcal{L}_{\mathrm{DSC}} + \lambda_{\mathrm{MTV}}\,\mathcal{L}_{\mathrm{MTV}} + \lambda_{\text{MTV-J}}\,\mathcal{L}_{\text{MTV-J}} \\ & + \lambda_{\mathrm{jac}}\,\mathcal{L}_{\mathrm{jac}} + \lambda_{\mathrm{TLG}}\,\mathcal{L}_{\mathrm{TLG}} + \lambda_{\mathrm{rigid}}\,\mathcal{L}_{\mathrm{rigid}} + \lambda_{\mathrm{smooth}}\,\mathcal{L}_{\mathrm{smooth}} + \lambda_{\mathrm{NDV}}\,\mathcal{L}_{\mathrm{NDV}}.\end{aligned} \tag{2}$$

All terms are evaluated on the composed transformation (1). Only the full-resolution level is trained with the complete objective. The two coarser levels use the similarity, label and regularisation terms alone, i.e. $\lambda_{\mathrm{MTV}} = \lambda_{\text{MTV-J}} = \lambda_{\mathrm{jac}} = \lambda_{\mathrm{TLG}} = \lambda_{\mathrm{rigid}} = 0$ at a quarter and half resolution, because the lesion and bone masks they would be evaluated on are only a few voxels wide there: after downsampling, thin ribs and small lesions are largely partial-volume, so both the volume ratios and the per-structure rigid fits measure interpolation rather than anatomy.

Registration Accuracy. Image similarity is measured by the local normalised cross correlation, evaluated as a multi-resolution sum over as many scales as the level has resolutions, i.e. one scale at the coarsest and three at the full-resolution level. The window is 5^3 voxels at the quarter-resolution level and 7^3 at the half- and full-resolution levels, so that the coarsest level, where one voxel already covers roughly a centimetre, is not asked to match structure over a neighbourhood far larger than the deformations it estimates. It is computed on the CT channel only ($\lambda_{\mathrm{PET}} = 0$): PET intensity is not conserved between the two time points, as lesions appear, resolve or change uptake under therapy, so a PET similarity term would reward matching disease rather than anatomy. Furthermore, we used a standard DSC loss for weak CT label supervision. Note that the objective contains no surface-distance term, although HD95 is scored; Sec. 3.2 shows that one would have nothing to improve.

Preservation of PET quantification. The scored MTV bias only constrains the *net* lesion volume, so we penalize it directly while supplying dense gradients through a Jacobian-based surrogate. All terms are evaluated on the total transformation $\varphi = \varphi_{\text{aff}} \circ \varphi_\theta$, i.e. the affine pre-registration composed with the predicted displacement, against the original moving lesion mask M, since the affine itself is generally not volume-preserving. First, we minimize the squared relative volume error

$$\mathcal{L}_{\text{MTV}} = \left(\frac{|M \circ \varphi| - |M|}{|M|} \right)^2, \tag{3}$$

where $M \circ \varphi$ denotes the linearly resampled mask. This matches the scored metric exactly, but its gradient is confined to the lesion's sub-voxel boundary shell. We therefore add a complementary term based on the change-of-variables identity: summing $\det J_\varphi$ over the warped lesion region recovers the original lesion volume, so the mean determinant over the warped lesion equals the ratio of original to warped volume. Driving this mean to one,

$$\mathcal{L}_{\text{MTV-J}} = \left(\frac{1}{|W|} \sum_{x \in W} \det J_\varphi(x) - 1 \right)^2, \tag{4}$$

where W denotes the warped lesion region, therefore pins the same net volume ratio as (3) while providing gradients throughout the lesion interior; unlike a per-voxel constraint, it still permits internal redistribution of volume. Finally, the per-voxel penalty

$$\mathcal{L}_{\text{jac}} = \frac{1}{|W|} \sum_{x \in W} \left(\det J_\varphi(x) - 1 \right)^2 \tag{5}$$

acts as a stricter local regulariser, discouraging compensating compression and expansion within the lesion.

The challenge additionally scores the total lesion glycolysis (TLG), the PET-activity-weighted counterpart of the MTV, which we penalize analogously as the relative change of intensity mass under the warp:

$$\mathcal{L}_{\text{TLG}} = \frac{\left| \sum_x (I \circ \varphi)(x)\, (M \circ \varphi)(x) - \sum_y I(y)\, M(y) \right|}{\sum_y I(y)\, M(y)}, \tag{6}$$

where I denotes the moving PET image. Since both the image and the mask are resampled with the same transformation, this term is sensitive not only to volume change but also to intensity blurring at the lesion boundary, and thus complements the purely geometric terms (3)-(5).

The same terms (3)-(6) are used during training and descended at test time by our instance optimisation (Sec. 2.4), so the network is trained on the objective it is later refined under.

Bone rigidity. In 146 patients with paired baseline and follow-up scans, a median of 46.0% (IQR 3.1 to 72.3%) of total lesion volume at baseline lay within skeletal structures, rising to 63.8% (IQR 12.1 to 77.4%) at follow-up. Since a substantial fraction of lesion volume is therefore intraosseous, a rigidity penalty on a bone mask is an anatomically grounded surrogate for the biomarker terms: enforcing rigid bone motion is both physiologically correct and, where lesions sit in bone, keeps the deformation locally volume-preserving. A rigid motion has $\det J_\varphi = 1$ and resamples the PET signal without compressing or expanding it, so it protects both scored biomarkers at once, the lesion volume that MTV measures and the intensity mass that TLG measures.

We first used the established local rigidity penalty, which combines deviation of the Jacobian J_φ from orthogonality [16, 23] with local affineness and orientation preservation [25, 19]. Its terms are evaluated by finite differences at every voxel and only then masked, so the stencil also reads non-bone voxels and, on thin cortical bone and ribs, ends up measuring the bone-to-soft-tissue displacement discontinuity that any correct registration must exhibit (Sec. 3.2). Eroding the mask would fix this, but at whole-body resolution many bone labels are only a few voxels wide.

We therefore replaced it with a per-structure rigid-fit residual over the 61 skeletal TotalSegmentator labels: for each structure we fit the best rigid transform to its own displacement in closed form by Kabsch alignment [12, 26] and penalise the mean squared residual, averaged within and then across structures. No stencil crosses the mask and each structure is fitted independently, so articulation is free; unlike the derivative-based penalties [25, 19] it constrains only the shape of the deformation and not its placement, and unlike [8] it needs no per-bone rigid field precomputed by a separate pre-registration, whose error would otherwise become a training target.

Deformation regularisation. To regularise the displacement field we used a diffusion regulariser, the mean squared spatial gradient of the displacement,

$$\mathcal{L}_{\text{smooth}} = \frac{1}{3} \sum_{d \in \{x,y,z\}} \frac{1}{|\Omega|} \sum_{p \in \Omega} \left\| \partial_d u(p) \right\|^2, \tag{7}$$

together with a term penalising non-diffeomorphic voxels. Rather than the usual penalty on negative Jacobian determinants of a single finite-difference stencil, we differentiably reproduce the scored metric itself: the challenge computes the non-diffeomorphic volume by decomposing each voxel into six tetrahedra and accumulating the negative part of their determinants over the body mask [15]. Writing $\det J_\varphi^{(k)}$ for the determinant of the k-th tetrahedral scheme, the loss

$$\mathcal{L}_{\text{NDV}} = \frac{100}{12|\Omega|} \sum_{p \in \Omega} \sum_{k=1}^{6} \max\bigl(-\det J_\varphi^{(k)}(p),\, 0\bigr) \tag{8}$$

equals the scored percentage of non-diffeomorphic volume and places gradients on exactly the folded voxels, so that minimising it minimises the reported %NDV directly.

2.4 Instance Optimisation

At test time we refine each pair individually. Keeping the predicted field fixed, we optimise a residual stationary velocity field, initialised to zero and exponentiated by scaling and squaring, on top of it, so the refined transformation stays diffeomorphic by construction and the network is only ever corrected, never overwritten. The objective is the same one the network was trained under, only reweighted so that the biomarker terms dominate. We descend it for 90 Adam steps with a learning rate of 10^{-2} and keep the best rather than the last iterate, scoring each step with the hard, nearest-neighbour-counted DSC, HD05, MTV and TLG instead of the bilinear proxies the optimiser differentiates, so the selected step is the one the scorer would prefer.

Submission container. Instance optimisation needs organ segmentations of both scans and lesion segmentations of the moving scan, and the submitted container has to produce them and register the pair within a budget of 90 s per image pair, which the configuration above exceeds. The deployed method therefore differs from the one evaluated in validation in three points, all of them inside instance optimisation or in the labels it consumes; everything up to and including the predicted field is identical. First, PET lesion masks come from an nnU-Net trained on this dataset with progressive patch-size growing [7], which is faster at inference than the ensemble used in validation. Second, CT organ labels are produced with TotalSegmentator in its fast mode together with its body segmentation, and only within a 100-slice axial band, selected as the region in which pre-optimisation Dice is worst and refinement therefore has the most to gain; segmenting the full volume does not fit the budget. Third, instance optimisation runs for 18 steps at a learning rate of 0.1 instead of 90 steps at 10^{-2}. We report both configurations separately in Sec. 3.1, since only the validation one is free of the time constraint.

2.5 Model Selection

Model selection enters our pipeline twice: once *within* a run, to decide which checkpoint to keep, and once *across* runs, to decide which variant to submit in the container. Both use the same three-component quality notion, and differ only in how each component is scored.

Both stages combine registration accuracy (CT DSC and HD95), PET-biomarker preservation (absolute MTV and TLG bias) and deformation regularity (%NDV after the official 0.005% practical-equivalence adjustment) as a weighted geometric mean with exponents 0.4, 0.4 and 0.2, mirroring the challenge's final score

$$S = 100 \cdot A^{0.4}\, B^{0.4}\, R^{0.2}, \tag{9}$$

where the accuracy term A is the geometric mean of the DSC and HD95 scores, the biomarker term B that of the MTV and TLG scores, and R the %NDV score.

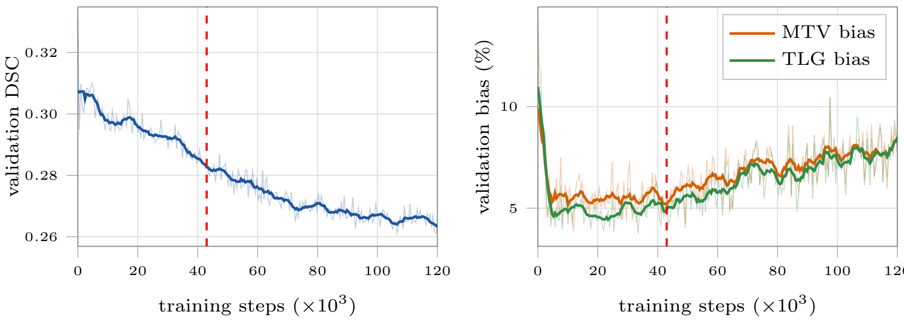

Fig. 2. Accuracy and biomarker preservation do not peak at the same checkpoint. Validation CT Dice (left) rises to a plateau, while the MTV and TLG bias (right) reach their optimum roughly halfway through level-3 training and degrade thereafter. The dashed line marks the checkpoint selected by Eq. (9). Faint lines are the per-round values, solid lines a centred rolling mean.

Checkpoint selection. Registration accuracy and tumour preservation do not peak together: in level-3 training, validation Dice improves until it plateaus, whereas the tumour metrics reach their optimum roughly halfway through and degrade thereafter (Fig. 2). Selecting on Dice alone therefore yields a checkpoint well past the tumour optimum, so we select on Eq. (9) instead. The official per-metric scores are ranks within a cohort of competing methods, which does not exist during a single training run. We therefore replace each rank score with a logistic quality centred on the metric's median and scaled by its robust spread (interquartile range divided by 1.349), both measured over the validation rounds of a completed run.

Submission selection. Choosing between trained variants does provide a cohort, so here we evaluate Eq. (9) exactly as the challenge does: each of the five metrics is ranked separately across the K variants, with ties taking the worst rank they span, and rank r is turned into the score $(K - r + 1)/K$. The variant with the highest S is the one we submit. To check that this choice is not an artefact of a few cases, we additionally compare the selected variant against every other variant per metric with a paired Wilcoxon signed-rank test over the validation cases, Holm-corrected within each metric. As the variant is selected on the same cases, these tests are exploratory rather than confirmatory; the official validation leaderboard serves as the held-out check.

2.6 Baselines

We compare against two references and three competing methods, all evaluated on the same pairs and under the identical protocol of Sec. 3.1. The two references bound the range in which a deformable method has to be useful: the unregistered pair, i.e. the identity transform, and our affine pre-registration alone, which is also the initialisation every method below starts from.

As a classical iterative baseline we ran NiftyReg [18, 17], using `reg_aladin` for the affine stage followed by `reg_f3d` initialised with it, so that the resulting control point grid encodes the total transform; all parameters were left at their defaults. As a learning-free optimisation baseline we ran ConvexAdam [?], in the form provided by the challenge organisers as the official baseline of the task, so that our numbers are directly comparable to the reference entry on the leaderboard. Finally, we compare against vanilla LapIRN, i.e. the same architecture and training schedule as our model but with the original objective only, without the DSC, NDV loss, PET-biomarker terms and the bone-rigidity penalty, which isolates the contribution of the proposed losses from that of the backbone.

2.7 Augmentation

With 239 training pairs, augmentation matters, and we chose it to reproduce the specific ways in which two sessions of the same patient differ rather than to maximise variety. Every transformation is applied consistently to both scans of a pair, to their labels and to the cached affine field, except where stated otherwise.

We flip left–right with probability 0.5, the only spatial flip that leaves whole-body anatomy plausible. On CT we apply an intensity shift of ± 0.02 in the normalised window (about ± 50 HU) and a scaling of 0.9 to 1.1, covering scanner and reconstruction variation. On PET we apply a scaling of 0.85 to 1.15 but no shift: SUV depends on the injected dose, the uptake time and the patient's weight, all of which differ between sessions, so the network must not key on absolute uptake, whereas an additive offset has no physical counterpart in an SUV map.

The last augmentation targets the difference the challenge singles out as the hardest. We crop randomly along the cranio-caudal axis, removing up to 40 slices from either end of both scans, which varies the extent of the acquired volume. On top of that we crop the moving and the fixed scan independently by up to a further 10 slices each, which is the part that actually reproduces a baseline/follow-up mismatch: it makes the two fields of view disagree, so the network is forced to register anatomy that is present in one scan and truncated in the other.

2.8 Training Setup

Each pyramid level is trained in sequence with Adam on whole volumes rather than patches, at a batch size of one with gradients accumulated over four steps. The convolutional trunk runs in `bfloat16` while the transformations and all losses are kept in single precision, since scaling and squaring and the Jacobian determinants compound rounding error. Every level starts with a five-epoch linear learning-rate warmup and keeps the preceding level frozen for its first ten epochs. The three levels are trained for 100k, 100k and 140k steps at learning rates of $3 \cdot 10^{-4}$, $2 \cdot 10^{-4}$ and $2.5 \cdot 10^{-4}$. The loss weights were chosen by the sweep of Sec. 3.1 and are given in the released code.

2.9 Computing Infrastructure

All models were trained and evaluated on a single NVIDIA H100 80 GB HBM3 (driver 610.57.04). The pipeline is implemented in Python 3.11 with PyTorch 2.6.0 [21] and CUDA 12.4; the affine pre-registration uses ANTsPy 0.6.3 [2]. Image and array handling rely on NiBabel 5.4.2, NumPy 2.4.4 and SciPy 1.17.1. Training all three levels takes [TODO: n] h, and inference is [TODO: n] s per pair, rising to [TODO: n] s with instance optimisation. Code and the submission container are available at [TODO: repository URL].

3 Results

3.1 Evaluation Protocol

The challenge does not release labels for the validation cases. Unless stated otherwise, the validation numbers we report are therefore computed against surrogate labels we generated ourselves: CT organ labels with TotalSegmentator and PET tumour labels with the winning autoPET-III submission. These surrogate labels underlie every validation metric in this work, including the checkpoint and submission selection of Sec. 2.5. Being predictions rather than the challenge's reference annotations, they carry their own error, so their absolute values are not directly comparable to the official validation leaderboard; we use them to compare methods against each other under an identical protocol. Where we report official validation-leaderboard results, these are computed by the organisers against their reference annotations and are marked as such. Since the leaderboard covers the same cases with the correct labels, we use the agreement between the two rankings as a check on the surrogate protocol, and prefer the leaderboard whenever it is available for the models being compared. We report our method in three configurations. *Ours (container)* is the submitted method of Sec. 2.4, run on a single NVIDIA H100 within the 90 s per image pair the challenge allows; since it is the configuration the hidden test set will be scored on, it is the one we test the baselines against. *Ours (validation)* is the unconstrained method of Sec. 2.2, reported as the upper bound that the time budget gives up; the two differ only in instance optimisation and in how the segmentations it consumes are produced. *Ours (no IO)* is the backbone alone, which isolates what test-time refinement contributes. The latter two carry no significance marks.

3.2 Ablation of the Objective

Derivative-based rigidity is ill posed on thin bone. We quantified the stencil leakage of Sec. 2.3 across five cases: 53.0% (range 51.6 to 54.7) of the voxels entering the loss were not bone, and 44.8% of the gradient magnitude fell outside bone. A field with zero displacement inside every bone, which is perfectly rigid by construction and should therefore cost nothing, scored 56 (range 40 to 69), all of which vanished after a single mask erosion. The term was thus measuring the bone-to-soft tissue displacement discontinuity that any correct registration must exhibit. With the per-structure rigid-fit penalty of Sec. 2.3, the zero displacement field now scores exactly 0 and a per-bone rigid field 2.4e-6, while anisotropic squeeze is still rejected at 1.16.

A surface-distance loss would not help. Since HD95 is scored, a differentiable surface-distance term such as the distance-transform Hausdorff surrogate of Karimi and Salcudean [13] is the obvious addition to the objective; we deliberately use none, in training or in instance optimisation. We analysed the 2296 scored (case, label) entries of the validation predictions before instance optimisation, so that the conclusion applies to the training objective and not only to the refined field. The median per-label HD95 is 3.27 mm and the 75th percentile 5.47 mm, i.e. exactly one and two voxels: three quarters of all labels already lie within two voxels of the discretisation floor attainable by a nearest-neighbour–warped label map. The mean is instead driven almost entirely by a small tail, with the worst 10% of entries carrying 48.1% of the total HD95 mass. Splitting that mass against the unregistered pair shows where it comes from: 48.1% is *inherent*, already bad before registration and no better after (mean 33.8 mm over 229 entries), 51.7% belongs to labels our field *improved* to a mean of 4.02 mm, and 0.0% was *created* by the registration. The inherent entries are dominated by structures for which no boundary refinement is meaningful: limbs subject to inter-session repositioning (humeri, femora, clavicles, together the largest single contribution), hollow organs whose filling changes between sessions and which therefore have no well-defined point correspondence (colon, small bowel, stomach, bladder, gallbladder), and anatomy truncated by the field of view (brain, skull), some of it reduced to a handful of voxels. A boundary loss would therefore expend its gradient on labels that are already at the floor, while leaving the tail - a capture-range and correspondence problem rather than a boundary-precision one - untouched.

3.3 Comparison against Baselines

We swept the loss weights and the instance-optimisation settings and selected the one reported below with the challenge ranking of Sec. 2.5, applied to the surrogate protocol above. Table 1 places the three configurations against the baselines. The container configuration improves DSC from 73.3% (NiftyReg, the strongest baseline) to 88.3% and halves HD95 from 7.09 to 4.77 mm, while reducing the MTV and TLG errors by more than a factor of two and the non-diffeomorphic volume from 1525 to 21 ppm. All differences are significant on every metric (paired Wilcoxon signed-rank, Holm-corrected, $p < 0.01$).

Table 1. Our method against the baselines on the official challenge validation set ($n = 20$ cases), evaluated against the surrogate labels of Sec. 3.1. *Ours (container)* is the submitted method of Sec. 2.4 run on a single NVIDIA H100 and within the 90 s per image pair the challenge allows and the configuration the baselines are tested against; *Ours (validation)* is the unconstrained configuration, reported as the upper bound the container's 90 s per-pair budget gives up; *Ours (no IO)* is the backbone without instance optimisation. Each cell reports the mean over cases. DSC and the MTV/TLG errors are given in %, NDV in ppm. Time is the measured wall-clock runtime for one image pair on the hardware of Sec. 2.9; it is a single measurement, not a per-case distribution, and is not tested or marked. Bold marks the best value in each column. * marks metrics on which *Ours (container)* is significantly better than the baseline in the given row (paired Wilcoxon signed-rank, Holm-corrected within each metric, $p < 0.05$); the other two rows of ours are reported for reference and carry no significance marks. Dashes mark metrics that are undefined for a baseline: the unregistered and affine cases have no deformation field, and the PET biomarkers are only meaningful after resampling.

Method	DSC (%) ↑	HD95 (mm) ↓	MTV %err ↓	TLG %err ↓	NDV (ppm) ↓	Time (s) ↓
Initial	14.8*	30.9*	–	–	–	–
Affine	52.9*	11.7*	5.47	4.70	–	–
NiftyReg	73.3*	7.09*	4.46	8.68*	1525	–
ConvexAdam	63.0*	8.41*	5.10	4.15	100	–
Ours (no IO)	76.5	6.65	3.22	5.08	**0.00**	–
Ours (container)	79.4	6.21	3.63	2.86	6633	–
Ours (validation)	**88.3**	**4.77**	**1.83**	**1.80**	20.7	–

3.4 Validation Leaderboard

3.5 Qualitative Results

4 Discussion

[TODO: discussion + short conclusion]

Acknowledgments. [TODO:]

Disclosure of Interests. [TODO:]

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FIGURE 5 – Standard qualitative panel: fixed, moving, warped and difference image, plus a $\det J_\varphi$ *heat map and a zoom on an intraosseous lesion showing that the lesion volume is preserved. Include one failure case (bowel or bladder filling) – the 2025 papers that showed a failure case read as more honest.*

Fig. 3. Qualitative registration results. [TODO: caption]

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